

```
$ sudo apt-get update
$ sudo apt-get install ubuntu-zfs
```

Once the installation is over, run the command shown below:

```
$ sudo zfs list
```

This should give an output that states 'No datasets available', which ensures that the installation is fine.

Now we have just one hard drive.

Configuring ZFS

As mentioned earlier, our installation has just one hard drive. Let's suppose that we need to add six more hard drives to our system.

Now, let us assume that we have finished adding the six hard drives. But since none of them is partitioned, they are currently unusable.

This is where the benefits of ZFS come in, freeing us from the burden of creating partitions (although we have the liberty of creating these, if we want to).

Let's now create a storage pool using two of our hard drives:

```
$ sudo zpool create -f <storage pool name> </path/to/driver1>
</path/to/driver2>
```

`zpool create` is the command used to create a new storage pool, and `-f` overrides any errors that occur (such as if the disk(s) already have information on them).

To see the newly created pool, run the following command:

```
$ sudo zfs list
```

Now let us look at where RAID levels come into play. What if we create a 2KB file in the storage pool that we just made? 1KB would automatically go to the first drive and 1KB to the second one. Then when we read the 2KB file, each hard drive would present 1KB to us, combining the speed of the two drives. Here we have used *RAID 0*. When you use *RAID 0*, your multiple disks appear to be a larger and faster hard disk. One disadvantage of *RAID 0* is that it is fragile. If any one of the drives dies, we lose our entire data. But ZFS supports higher RAID levels that can overcome this disadvantage.

To get a feel of how ZFS overcomes data loss as in the above case, let us look at another scenario.

First, let's delete the pool that we have created, as follows:

```
$ sudo zpool destroy <storage pool name>
```

Now we are going to create a RAID Z pool, which is an improved version of a RAID 5 pool (RAID 5 needs at least

three drives and uses striping to divide data across all the hard drives, with additional parity data divided across all disks. If one of the hard drives dies, you won't lose any of your data. RAID 5 offers data redundancy with a lower storage cost than RAID 1). RAID Z voids the 'write hole' by using copy-on-write. If a single disk in your pool dies, simply replace that disk, and ZFS will automatically rebuild the data based on parity information from the other disks. To lose all the information in your storage pool, two disks would have to die. To make things even more redundant, you can use RAID 6 (RAID-Z2 in the case of ZFS) and have double parity.

To achieve the above, run the following command:

```
$ sudo zpool create -f <storage pool name> raidz <driver1>
<driver2> <driver3>
```

What makes ZFS different?

ZFS is significantly different from any previous file system because it is more than just a file system. Combining the traditionally separate roles of volume manager and file system provides ZFS with unique advantages. The file system is now aware of the underlying structure of the disks. Traditional file systems could only be created on a single disk — one at a time. If there were two disks, then two separate file systems would have to be created. In a traditional hardware RAID configuration, this problem was avoided by presenting the operating system with a single logical disk made up of the space provided by a number of physical disks, on top of which the operating system placed a file system. ZFS's combination of the volume manager and the file system solves this, and allows the creation of many file systems — all sharing a pool of available storage.

One of the biggest advantages of ZFS's awareness of the physical layout of the disks is that existing file systems can be grown automatically when additional disks are added to the pool. This new space is then made available to all the file systems. ZFS also has a number of different properties that can be applied to each file system, giving many advantages to creating a number of different file systems and datasets rather than a single monolithic file system. **END** 

References

- [1] <https://www.freebsd.org/doc/handbook/zfs.html>
- [2] <http://www.howtogeek.com/175159/an-introduction-to-the-z-file-system-zfs-for-linux/>

By: Athira Lekshmi C.V.

The author is pursuing a Bachelor's degree in computer science and engineering at Amrita Vishwa Vidyapeetham in Kollam, Kerala. She is a FOSS enthusiast and an active member of FOSS club in her college.